

Python Lab — Smart HealthCare Analytics

BSDSF24A

Time: 1.5 hrs.

Marks: 100.

Case Study

The National HealthCare Innovation Center (NHIC) wants to improve patient care through data analytics. As a Junior Data Analyst, your responsibility is to analyze patient vitals, identify risk cases, and generate insights.

Dataset Format

```
patients_data = [  
    "id:1,name:Ali Khan,age:45,doctor:Dr.  
    Ahmed,hr:88,bp:120/80,glucose:110,feedback:Good",  
    "id:2,name:Ayesha Noor,age:63,doctor:Dr.  
    Sara,hr:102,bp:145/95,glucose:160,feedback:Critical",  
    ...]
```

Q1 – Patient Record Conversion (Dictionary Format)

Explanation:

In this question, you must convert each raw text string into a structured dictionary. The raw string contains multiple fields separated by commas. Each field has a key and value separated by a colon (:). Your task is to:

- Split the string using commas to isolate each field.
- Split each field using ':' to separate key and value.
- Convert numeric fields (id, age, hr, glucose) into integers.
- Store everything in a dictionary.
- Append all dictionaries into a list.

Final dictionary structure example:

```
{  
    "id": 1,  
    "name": "Ali Khan",  
    "age": 45,
```

```
"doctor": "Dr. Ahmed",  
"hr": 88,  
"bp": "120/80",  
"glucose": 110,  
"feedback": "Good"  
}
```

Q2 – Count Total Patients

Count how many patients exist in the dataset.

Q3 – Average Heart Rate Per Doctor

Calculate average heart rate grouped by doctor.

Q4 – High-Risk Patients (HR > 100)

List all patients whose heart rate is above 100.

Q5 – Lowest Glucose Level

Find the patient having the lowest glucose reading.

Q6 – Highest Glucose Level

Find the patient with the highest glucose reading.

Q7 – Blood Pressure Alert (BP > 140/90)

Return all patients whose BP is above 140/90.

Q8 – Distinct Doctors Count

Count the total number of unique doctors.

Q9 – Feedback Distribution: Count how many patients fall under each feedback type (Excellent, Good, Fair, Critical).

Q10 – Overall Average Heart Rate

Calculate average HR across all patients.

Q11 – Health Grade Assignment Assign health grade:

- $HR < 70 = A$ (Excellent)
- $HR 70-89 = B$ (Stable)
- $HR 90-110 = C$ (Concern)
- $HR > 110 = D$ (High Risk)

Q12 – Glucose Improvement (–5%)

Assume glucose is reduced by 5% for every patient.

Calculate updated glucose values.

Q13 – Doctor Summary

For each doctor show:

- number of assigned patients
- average HR
- average glucose

Q14 – Senior Patients (Age ≥ 60)

List all senior patients.

Q15 – Critical Condition Tag

Mark patients with either:

- $BP > 140/90$
- OR feedback = "Critical"

As "Severe", else "Normal"

Q16 – Sort By Age

Sort patients from oldest to youngest.

Q17 – Underrated Cases

Patients having normal HR (<90) but feedback "Fair" or "Critical".

Q18 – Lowercase Feedback + Count 'critical'

Convert all feedback to lowercase and count how many are “critical”.

Q19 – Doctor Load Check

Verify each doctor has at least 2 patients.

Return list of overworked or understaffed doctors.

Q20 – Patient Summary Card

Print a one-line summary:

Ali Khan | 45 yrs | Dr. Ahmed | HR:88 | BP:120/80 |
Glucose:110 | Feedback:Good